BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm | Semester: Fall 2025
Duration: 1 hour 30 min | Full Marks: 40

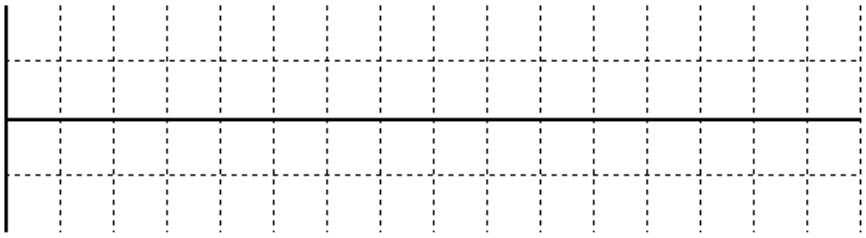
CSE 320/EEE361/ECE361: Data Communications

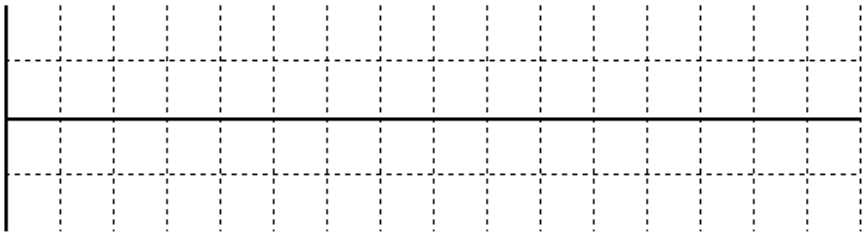
Answer the following questions.
Figures in the right margin indicate marks.

SET A

1. [CO1]	a)	Innovate Corp is a medium-sized company expanding into a new two-story headquarters. The company has tasked you with designing the **network topology** for two key areas. **Floor 1:** The Executive & Finance Wing This floor houses the CEO, CFO, finance server and backup server. The primary requirement is maximum reliability and fault tolerance. If one connection fails, it should not disrupt communication between any other devices on this part of the network. **Floor 2:** The Media Production Lab This lab contains 5 high-powered workstations used for video rendering. The data flow in one direction; one workstation processes a segment of a project and passes it to the next in the pipeline in clockwise fashion. The two floors are connected via a backbone cable providing high data rate. Your Task: Based on the scenarios and hints above, **Identify** and **design** the network topology. **Determine** the total number of links in the network.	[1+2+2]
	b)	**Identify** the layers of **OSI model** which are responsible for the phrases in bold: During a remote surgery session, a robotic arm in a hospital's operation theater is controlled by a surgeon located thousands of kilometers away. The surgeon's console transmits **instruction messages** [1] to the robotic system through the internet. These commands are encapsulated within encrypted packets that include **logical addressing** [2] to identify both the surgeon's console and the hospital's server. The hospital's router segments the data stream into **smaller frames** [3] and adds physical addresses to each before transmitting them over a **fiber-optic network** [4]. Upon reaching the	[3]

		hospital network, the server reassembles the frames, decrypts the data, and **acknowledges successful receipt of the instructions** [5]. During the operation, a slight delay occurs, prompting the system to request **retransmission of a lost segment** [6]. Despite this, the connection remains stable, and video feedback from the camera feed is rendered in real-time on the surgeon's console.	
	c)	What is Jitter and how does it affect the effectiveness of a network system? Explain why high jitter would be particularly disruptive for a live video conferencing call but less so for downloading an email.	[2]
	d)	What are the key benefits of using a layered model in a data communication system?	[2]
2. [CO2]	a)	A transmission line is 20 km long, where A is the source and D is the destination. To prevent excessive signal degradation, an amplifier is placed between point B and C. As a result, the received signal at point C gets amplified by 5 times than the signal received at point B. Point B is located 9 km away from the source. The signal power measured at the destination D is 20 W. The attenuation along the segment C to D is 6.80 dB. **Calculate** the signal power at point B. **[Hint. There are two segments in the overall transmission medium]**	[4]
	b)	You are transmitting a periodic composite signal to a friend who is **500 km** away through a communication channel. The signal can be decomposed into **four** sine waves with time periods of 25 ms, 35 ms, 60 ms, and 10 ms respectively. Each sine wave has a peak amplitude of 15 V. **Draw** the frequency-domain representation (spectrum) of the composite analog signal.	[2]
	c)	If the capacity of a noisy channel is needed to be increased, what changes or improvements need to be made to the channel?	[2]
	d)	Suppose, a document file containing 6 pages is being transmitted over a noiseless channel at a speed of 160 kbps. Each page contains 70 characters, and the file is encoded using ASCII (8 bits per character). The channel being used supports 29 signal levels. i. **Calculate** the bandwidth of the channel in Hz. If the channel is a low-pass channel, **determine** its highest frequency. ii. If the actual capacity of the channel is 35% less than the theoretical noiseless channel's capacity, **identify** the Signal-to-Noise Ratio (SNR) in decibels (dB). iii. **Compute** the transmission delay for sending the entire document file over this channel.	[3+3+1]

3. [CO2]	a)	A measurement system records a composite analog signal that is the sum of four sinusoidal components with frequencies and amplitudes as follows: 4 Hz (7 V), 20 Hz (2 V), 15 Hz (3 V), and 10 Hz (5 V). The signal is to be digitized using a PCM system that employs 256 quantization codes (levels). i. **Determine** the minimum sampling frequency required for proper construction. ii. **Identify** the number of bits per sample and the resulting **minimum bit rate** (in bits per second) of the PCM bitstream.	[2+2]
	b)	You are designing a custom 3B/5B block encoding scheme for a digital communication system. Each 3-bit data block will be substituted by a 5-bit codeword to improve clock recovery and limit long runs of zeros. A valid 5-bit codeword must **start with 1, contain no '000', have at most three zeros in total, and must not end with '00'. Four** of the codewords are already provided as reference in the partial mapping table below: Dataword / Codeword 000 / 001 / 11101 010 / 011 / 11010 100 / 101 / 10110 110 / 10111 111 / Fill in the missing **5-bit Codewords** for the remaining Dataword using valid codewords that follow all the rules given above. ***(Write the completed table on your question.)***	[4]
	c)	Suppose in a system, a multi-level line coding scheme will be used with the following properties: There will be synchronization issues and a DC problem for a long sequence of 0's, but a long sequence of 1's does not contribute to any such problems. **Which** line coding technique will be followed? Now, **encode** the following binary bits **using the graph below**: **110000101011** 	[2]

	d)	Apply a technique that can resolve both the synchronization issues and the DC problem for a long sequence of 0's without requiring any extra bandwidth. **What** is the name of the technique? **Encode** the following binary bits **using the graph below**: **11000000010** 	[3]

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